

# Lab Assignment 07



Inspiring Excellence

|                  |                                   |
|------------------|-----------------------------------|
| Course Code:     | CSE111                            |
| Course Title:    | Programming Language II           |
| Topic:           | Static variable and Static Method |
| Number of Tasks: | 10 (Classwork: 05, Homework: 05)  |

*[Submit all the Coding Tasks (Homework: Task 1 to 5) in the Google Form shared on buX before the next lab.]*

[You are not allowed to change the driver codes of any of the tasks]

## CLASSWORK

### Task 1

Design the **Passenger** class in such a way that the following code provides the expected output.

- Passenger class has two static variables *no\_of\_passenger* and *total\_fare*.
- Each passenger has to pay 20 TK/Distance and extra 10 TK/BaggageWeight.

| Given Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Expected Output                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>public class PassengerTester{     public static void main(String args[]){         System.out.println("Total Passenger: " + Passenger.no_of_passenger);         System.out.println("Total Fare: " + Passenger.total_fare + " TK");         System.out.println("=====1=====");         Passenger p1 = new Passenger("Lara", 5.6);         p1.passengerDetails();         System.out.println("=====2=====");         Passenger p2 = new Passenger("Kevin", 10.0);         p2.storeBaggageWeight(6.8);         p2.passengerDetails();         System.out.println("=====3=====");         Passenger p3 = new Passenger("Robin", 2.3);         p3.storeBaggageWeight(5.0);         p3.passengerDetails();         System.out.println("=====4=====");         System.out.println("Total Passenger: " + Passenger.no_of_passenger);         System.out.println("Total Fare: " + Passenger.total_fare + " TK");     } }</pre> | <pre>Total Passenger: 0 Total Fare: 0.0 TK =====1===== Name: Lara Fare: 112.0 TK =====2===== Name: Kevin Fare: 268.0 TK =====3===== Name: Robin Fare: 96.0 TK =====4===== Total Passenger: 3 Total Fare: 476.0 TK</pre> |

## Task 2

Design a **Book** class in such a way that the following code provides the expected output.

- The Book class has two static variables: total\_books\_sold and total\_revenue.
- Each book has a base price of 150 TK. If the discountPercentage is applied, the book's price is reduced by that percentage.
- The Book class should have a method to calculate the price after the discount

| Given Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Expected Output                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>public class BookTester {     public static void main(String[] args) {         System.out.println("Total Books Sold: " + Book.total_books_sold);         System.out.println("Total Revenue: "+Book.total_revenue + " TK");         System.out.println("=====1=====");          Book b1 = new Book("Java Programming", 10); // 10% discount         b1.bookDetails();          System.out.println("=====2=====");          Book b2 = new Book("Python Programming", 15); // 15% discount         b2.bookDetails();          System.out.println("=====3=====");          Book b3 = new Book("Data Structures", 5); // 5% discount         b3.bookDetails();          System.out.println("=====4=====");         System.out.println("Total Books Sold: " + Book.total_books_sold);         System.out.println("Total Revenue: "+Book.total_revenue + " TK");     } }</pre> | <pre>Total Books Sold: 0 Total Revenue: 0.0 TK =====1===== Title: Java Programming Price after Discount: 135.0 TK =====2===== Title: Python Programming Price after Discount: 127.5 TK =====3===== Title: Data Structures Price after Discount: 142.5 TK =====4===== Total Books Sold: 3 Total Revenue: 405.0 TK</pre> |

### Task 3

Design a **Student** class in such a way that the following code provides the expected output.

| Driver Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>public class StudentTester {     public static void main(String[] args) {         Student.printDetails();         System.out.println("-----");         Student mikasa = new Student("Mikasa", 3.75);         mikasa.individualDetail();         System.out.println("-----");         Student.printDetails();         System.out.println("-----");         Student harry = new Student("Harry", 2.5, "Charms");         harry.individualDetail();         System.out.println("-----");         Student.printDetails();         System.out.println("-----");         Student levi = new Student("Levi", 3.33);         levi.individualDetail();         System.out.println("-----");         Student.printDetails();     } }</pre> | <pre>Total Student(s): 0 CSE Student(s): 0 Other Department Student(s): 0 ----- ID: 1 Name: Mikasa CGPA: 3.75 Department: CSE ----- Total Student(s): 1 CSE Student(s): 1 Other Department Student(s): 0 ----- ID: 2 Name: Harry CGPA: 2.5 Department: Charms ----- Total Student(s): 2 CSE Student(s): 1 Other Department Student(s): 1 ----- ID: 3 Name: Levi CGPA: 3.33 Department: CSE ----- Total Student(s): 3 CSE Student(s): 2 Other Department Student(s): 1</pre> |

## Task 4

Suppose you have opened a new library, from where your friends can borrow books. Initially you have bought 3 books (Pather Panchali, Durgesh Nandini & Anandmath) each of 3 copies only. Design the **Borrower** class in such a way that the following code provides the expected output.

- You are given the arrays *book\_count* and *book\_name* to keep track of the number of books available. For simplicity, assume that there will be no other books in the library.
- You must reuse the *remainingBooks()* method when needed.

| Given Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Expected Output                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>public class Tester{     public static void main(String args[]){         Borrower.bookStatus();         System.out.println("*****1*****");         Borrower b1 = new Borrower("Nabila");         b1.borrowBook("Pather Panchali");         b1.borrowBook("Anandmath");         b1.borrowerDetails();         System.out.println("*****2*****");         Borrower b2 = new Borrower("Sadia");         b2.borrowBook("Anandmath");         b2.borrowBook("Durgesh Nandini");         b2.borrowBook("Pather Panchali");         b2.borrowerDetails();         System.out.println("*****3*****");         System.out.println(Borrower.remainingBooks("Anandmath")+         "copies of Anandmath is remaining.");         System.out.println("*****4*****");         Borrower b3 = new Borrower("Anika");         b3.borrowBook("Anandmath");         Borrower.bookStatus();         System.out.println("*****5*****");         Borrower b4 = new Borrower("Oishi");         b4.borrowBook("Anandmath");         b4.borrowBook("Durgesh Nandini");         b4.borrowerDetails();     } }  public class Borrower{     public static int book_count[] = {3, 3, 3};     public static String book_name[] = {"Pather Panchali",     "Durgesh Nandini", "Anandmath"};      // Your Code here }</pre> | <pre>Available Books: Pather Panchali: 3 Durgesh Nandini: 3 Anandmath: 3 *****1***** Name: Nabila Books Borrowed: Pather Panchali Anandmath *****2***** Name: Sadia Books Borrowed: Anandmath Durgesh Nandini Pather Panchali *****3***** 1 copies of Anandmath is remaining. *****4***** Available Books: Pather Panchali: 1 Durgesh Nandini: 2 Anandmath: 0 *****5***** This book is not available. Name: Oishi Books Borrowed: Durgesh Nandini</pre> |

## Task 5

|    |                                             |
|----|---------------------------------------------|
| 1  | public class B{                             |
| 2  | public static int x;                        |
| 3  | public int y = 4;                           |
| 4  | public int temp = -5;                       |
| 5  | public int sum = 2;                         |
| 6  | public B(){                                 |
| -  | y = temp + 3 ;                              |
| 8  | sum = 3 + temp + 3;                         |
| 9  | temp-=2;                                    |
| 10 | }                                           |
| 11 | public B(B b){                              |
| 12 | sum = b.sum;                                |
| 13 | x = b.x;                                    |
| 14 | b.methodB(1,3);                             |
| 15 | }                                           |
| 16 | public void methodA(int m, int n){          |
| 17 | int x = 2;                                  |
| 18 | y = y + m + (temp++);                       |
| 19 | x = x + 7 + n;                              |
| 20 | sum = sum + x + y;                          |
| 21 | System.out.println(x + " " + y+ " " + sum); |
| 22 | }                                           |
| 23 | public void methodB(int m, int n){          |
| 24 | int y = 0;                                  |
| 25 | y = y + this.y;                             |
| 26 | x = this.y + 3 + temp;                      |
| 27 | methodA(x, y);                              |
| 28 | sum = x + y + sum;                          |
| 29 | System.out.println(x + " " + y+ " " + sum); |
| 30 | }                                           |
| 31 | }                                           |

**Consider the following code:**

|                   |   |   |     |
|-------------------|---|---|-----|
| B b1 = new B();   | x | y | sum |
| B b2 = new B(b1); |   |   |     |
| b1.methodA(3, 2); |   |   |     |
| b2.methodB(1, 2); |   |   |     |
|                   |   |   |     |
|                   |   |   |     |

# HOMEWORK

## Task 1

Design the **Player** class with the necessary property to produce the output from the given driver code. **Hint: The total number of players is maximum 11**

| Driver Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>public class PlayerTester{     public static void main(String[] args) {         System.out.println("Total number of players: " +         Player.total);         System.out.println("1-----");         Player p1 = new Player("Neymar", "Brazil",5);         System.out.println(p1.player_detail());         System.out.println("=====");         Player.info();         System.out.println("2-----");         Player p2 = new Player("Ronaldo", "Portugal", 7);         System.out.println(p2.player_detail());         System.out.println("=====");         Player.info();         System.out.println("3-----");         Player p3 = new Player("Messi", "Argentina", 6);         System.out.println(p3.player_detail());         System.out.println("=====");         Player.info();         System.out.println("4-----");         Player p4 = new Player("Mbappe", "France", 10);         System.out.println(p4.player_detail());         System.out.println("=====");         Player.info();     } }</pre> | <pre>Total number of players: 0 1----- Player Name: Neymar Jersey Number: 5 Country: Brazil ===== Total number of players: 1 Players enlisted so far: Neymar 2----- Player Name: Ronaldo Jersey Number: 7 Country: Portugal ===== Total number of players: 2 Players enlisted so far: Neymar, Ronaldo 3----- Player Name: Messi Jersey Number: 6 Country: Argentina ===== Total number of players: 3 Players enlisted so far: Neymar, Ronaldo, Messi 4----- Player Name: Mbappe Jersey Number: 10 Country: France ===== Total number of players: 4 Players enlisted so far: Neymar, Ronaldo, Messi, Mbappe</pre> |

## Task 2

Implement the design of the **Travel** class so that the following output is produced. Use Encapsulation to protect the variables. [Hint: Assign all the variables in private]

| Driver Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Output                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>public class TravelTester {     public static void main(String[] args) {         System.out.println("No. of Traveller = " + Travel.getCount());         System.out.println("1=====");          Travel t1 = new Travel("Dhaka", "India");         System.out.println(t1.displayTravelInfo());         System.out.println("2=====");          Travel t2 = new Travel("Kuala Lumpur", "Dhaka");         t2.setTime(23);         System.out.println(t2.displayTravelInfo());         System.out.println("3=====");          Travel t3 = new Travel("Dhaka", "New_Zealand");         t3.setTime(15);         t3.setDestination("Germany");         System.out.println(t3.displayTravelInfo());         System.out.println("4=====");          Travel t4 = new Travel("Dhaka", "India");         t4.setTime(9);         t4.setSource("Malaysia");         t4.setDestination("Canada");         System.out.println(t4.displayTravelInfo());         System.out.println("5=====");          System.out.println("No. of Traveller = " + Travel.getCount());     } }</pre> | <pre>No. of Traveller = 0 1===== Source: Dhaka Destination: India Flight Time: 1:00 2===== Source: Kuala Lumpur Destination: Dhaka Flight Time: 23:00 3===== Source: Dhaka Destination: Germany Flight Time: 15:00 4===== Source: Malaysia Destination: Canada Flight Time: 9:00 5===== No. of Traveller = 4</pre> |



### Task 3

For this task, you need to design the **Cargo** class with appropriate static and non-static variables and methods to produce this given output for the given tester code.

**Note:** .load() method marks an object as selected for transport, and .unload() method unmarked it. At a time, the transport capacity is 10.0 tonnes. Each Cargo object is initialized with 2 attributes from the constructor - the contents and the weight. Carefully observe the outputs to identify the other attributes and design the class.

| Given Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Expected Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>public class CargoTester {     public static void main(String[] args) {         System.out.println("Cargo Capacity: "+ Cargo.capacity());         System.out.println("1=====");         Cargo a = new Cargo("Industrial Machinery", 4.5);         a.details();         System.out.println("2=====");         a.load();         System.out.println("3=====");         Cargo b = new Cargo("Steel Ingot", 2.7);         b.details();         System.out.println("4=====");         System.out.println("Cargo Capacity: "+ Cargo.capacity());         System.out.println("5=====");         b.load();         System.out.println("Cargo Capacity: "+ Cargo.capacity());         System.out.println("6=====");         Cargo c = new Cargo("Tree Trunks", 3.6);         c.load();         System.out.println("7=====");         c.details();         b.details();         System.out.println("8=====");         Cargo d = new Cargo("Processed Goods", 1.8);         d.load();         System.out.println("Cargo Capacity: "+ Cargo.capacity());         System.out.println("9=====");         b.unload();         System.out.println("Cargo Capacity: "+ Cargo.capacity());         System.out.println("10=====");         c.load();         System.out.println("11=====");         b.details();         System.out.println("Cargo Capacity: "+ Cargo.capacity());     } }</pre> | <pre>Cargo Capacity: 10.0 1===== Cargo ID: 1, Contents: Industrial Machinery, Weight: 4.5, Loaded: false 2===== Cargo 1 loaded for transport. 3===== Cargo ID: 2, Contents: Steel Ingot, Weight: 2.7, Loaded: false 4===== Cargo Capacity: 5.5 5===== Cargo 2 loaded for transport. Cargo Capacity: 2.8 6===== Cannot load cargo, exceeds weight capacity. 7===== Cargo ID: 3, Contents: Tree Trunks, Weight: 3.6, Loaded: false Cargo ID: 2, Contents: Steel Ingot, Weight: 2.7, Loaded: true 8===== Cargo 4 loaded for transport. Cargo Capacity: 1.0 9===== Cargo 2 unloaded. Cargo Capacity: 3.7 10===== Cargo 3 loaded for transport. 11===== Cargo ID: 2, Contents: Steel Ingot, Weight: 2.7, Loaded: false Cargo Capacity: 0.09999999999999964</pre> |

### Task 4

|    |                                             |                |  |  |
|----|---------------------------------------------|----------------|--|--|
| 1  | <b>public class FinalT6A{</b>               | <b>Outputs</b> |  |  |
| 2  | public static int temp = 3;                 |                |  |  |
| 3  | public int sum;                             |                |  |  |
| 4  | public int y = 2;                           |                |  |  |
| 5  | public FinalT6A(int x, int p){              |                |  |  |
| 6  | temp+=3;                                    |                |  |  |
| 7  | y = temp - p;                               |                |  |  |
| 8  | sum = FinalT6A.temp + x;                    |                |  |  |
| 9  | System.out.println(x + " " + y+ " " + sum); |                |  |  |
| 10 | }                                           |                |  |  |
| 11 | public void methodA(){                      |                |  |  |
| 12 | int x=0, y =0;                              |                |  |  |
| 13 | y = y + this.y;                             |                |  |  |
| 14 | x = this.y + 2 + temp;                      |                |  |  |
| 15 | sum = x + y + methodB(temp, y);             |                |  |  |
| 16 | System.out.println(x + " " + y+ " " + sum); |                |  |  |
| 17 | }                                           |                |  |  |
| 18 | public int methodB(int temp, int n){        |                |  |  |
| 19 | int x = 0;                                  |                |  |  |
| 20 | y = y + (++temp);                           |                |  |  |
| 21 | x = x + 2 + n;                              |                |  |  |
| 22 | sum = sum + x + y;                          |                |  |  |
| 23 | System.out.println(x + " " + y+ " " + sum); |                |  |  |
| 24 | return sum;                                 |                |  |  |
| 25 | }                                           |                |  |  |
| 26 | }                                           |                |  |  |
| 27 | <b>public class Test10{</b>                 |                |  |  |
| 28 | public static void main(String [] args){    |                |  |  |
| 29 | FinalT6A q1 = new FinalT6A(2,1);            |                |  |  |
| 30 | q1.methodA();                               |                |  |  |
| 31 | FinalT6A q2 = new FinalT6A(3,1);            |                |  |  |
| 32 | q2.methodB(7,8);                            |                |  |  |
| 33 | }                                           |                |  |  |
| 34 | }                                           |                |  |  |

## Task 5

|     |                                                 |               |  |  |
|-----|-------------------------------------------------|---------------|--|--|
| 1.  | <b>public class</b> Tracing {                   | <b>Output</b> |  |  |
| 2.  | <b>public static</b> int x= 0, y = 0;           |               |  |  |
| 3.  | <b>public</b> int a, b;                         |               |  |  |
| 4.  | <b>public</b> Tracing(int a, int b){            |               |  |  |
| 5.  | this.a = a;                                     |               |  |  |
| 6.  | this.b = b;                                     |               |  |  |
| 7.  | x+=1;                                           |               |  |  |
| 8.  | y+=2;                                           |               |  |  |
| 9.  | }                                               |               |  |  |
| 10. | <b>public void</b> methodA(int a){              |               |  |  |
| 11. | this.a = x+a;                                   |               |  |  |
| 12. | this.b = this.b+ this.a +this.methodB();        |               |  |  |
| 13. | System.out.println(this.a+" "+this.b+" "+x);    |               |  |  |
| 14. | }                                               |               |  |  |
| 15. | <b>public int</b> methodB(){                    |               |  |  |
| 16. | this.b = y - this.b + this.a;                   |               |  |  |
| 17. | System.out.println(this.a+" "+this.b+" "+x);    |               |  |  |
| 18. | x += this.b;                                    |               |  |  |
| 19. | return this.b;                                  |               |  |  |
| 20. | }                                               |               |  |  |
| 21. | <b>public void</b> methodB(Tracing t1){         |               |  |  |
| 22. | t1.b = this.y - t1.b + this.b;                  |               |  |  |
| 23. | System.out.println(t1.a+" "+t1.b+" "+x);        |               |  |  |
| 24. | }                                               |               |  |  |
| 25. | }                                               |               |  |  |
| 26. | <b>public class</b> Test9{                      |               |  |  |
| 27. | <b>public static void</b> main(String [] args){ |               |  |  |
| 28. | Tracing t1= new Tracing(2, 3);                  |               |  |  |
|     | t1.methodA(1);                                  |               |  |  |
| 29. | Tracing t2= new Tracing(3, 4);                  |               |  |  |
| 30. | t2.methodA(2);                                  |               |  |  |
| 31. | t1.methodB(t2);                                 |               |  |  |
| 32. | t2.methodB(t2);                                 |               |  |  |
| 33. | }                                               |               |  |  |
| 34. | }                                               |               |  |  |

## Ungraded Tasks (Optional)

(You don't have to submit the ungraded tasks)

### Task 1

Design the **SultansDine** class with the necessary property to produce the output from the given driver code.

Subtasks:

1. Create SultansDine class
2. Create 2 static variable and 1 static array
3. Create 1 static method
4. Calculation of branch sell is given below
  - a. If sellQuantity < 10:
    - i. Branch\_sell = quantity \* 300
  - b. Else if sellQuantity < 20:
    - i. Branch\_sell = quantity \* 350
  - c. Else
    - i. Branch\_sell = quantity \* 400
5. Calculation of branch's sell percentage = (branch's sell / total sell) \* 100

| Driver Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>public class SultansDineTester {     public static void main(String[] args) {         SultansDine.details();         System.out.println("1=====");         SultansDine dhanmondi = new SultansDine("Dhanmondi");         dhanmondi.sellQuantity(25);         dhanmondi.branchInformation();         System.out.println("2=====");         SultansDine.details();         System.out.println("3=====");         SultansDine baily_road = new SultansDine("Baily Road");         baily_road.sellQuantity(15);         baily_road.branchInformation();         System.out.println("4=====");         SultansDine.details();         System.out.println("5=====");         SultansDine gulshan = new SultansDine("Gulshan");         gulshan.sellQuantity(9);         gulshan.branchInformation();         System.out.println("6=====");         SultansDine.details();     } }</pre> | <pre>Total Number of branch(s): 0 Total Sell: 0 Taka 1===== Branch Name: Dhanmondi Branch Sell: 10000 Taka 2===== Total Number of branch(s): 1 Total Sell: 10000 Taka Branch Name: Dhanmondi, Branch Sell: 10000 Taka Branch consists of total sell's 100.00 3===== Branch Name: Baily Road Branch Sell: 5250 Taka 4===== Total Number of branch(s): 2 Total Sell: 15250 Taka Branch Name: Dhanmondi, Branch Sell: 10000 Taka Branch consists of total sell's 65.57 Branch Name: Baily Road, Branch Sell: 5250 Taka Branch consists of total sell's 34.43 5===== Branch Name: Gulshan Branch Sell: 2700 Taka 6===== Total Number of branch(s): 3 Total Sell: 17950 Taka Branch Name: Dhanmondi, Branch Sell: 10000 Taka Branch consists of total sell's 55.71 Branch Name: Baily Road, Branch Sell: 5250 Taka Branch consists of total sell's 29.25 Branch Name: Gulshan, Branch Sell: 2700 Taka Branch consists of total sell's 15.04</pre> |

## Task 2

| 1.  | <b>public class Maze{</b>                | <b>Output</b> |  |
|-----|------------------------------------------|---------------|--|
| 2.  | public static int x;                     |               |  |
| 3.  | public void methodA(){                   |               |  |
| 4.  | int m = 5;                               |               |  |
| 5.  | x=11;                                    |               |  |
| 6.  | System.out.println(x+" "+m);             |               |  |
| 7.  | m=methodB(m-3)+x;                        |               |  |
| 8.  | System.out.println(x+" "+(m));           |               |  |
| 9.  | methodB(x,m);                            |               |  |
| 10. | System.out.println(x+" "+m+x);           |               |  |
| 11. | }                                        |               |  |
| 12. | public int methodB(int y){               |               |  |
| 13. | x=y*y;                                   |               |  |
| 14. | System.out.println(x+" "+y);             |               |  |
| 15. | return x+3;                              |               |  |
| 16. | }                                        |               |  |
| 17. | public void methodB(int z, int x){       |               |  |
| 18. | z=z-2;                                   |               |  |
| 19. | x=x*1%z;                                 |               |  |
| 20. | System.out.println(z+" "+x);             |               |  |
| 21. | }                                        |               |  |
| 22. | }                                        |               |  |
| 23. | <b>public class TestU3{</b>              |               |  |
| 24. | public static void main(String [] args){ |               |  |
| 25. | Maze c = new Maze();                     |               |  |
| 26. | c.methodA();                             |               |  |
| 27. | c.methodB(-11, 45);                      |               |  |
| 28. | }                                        |               |  |
| 29. | }                                        |               |  |

### Task 3

Find the outputs after running the main() method in **TestU4** class.

| 1 public class Quiz1{                          | Outputs |  |  |
|------------------------------------------------|---------|--|--|
| 2 public static int temp = 4;                  |         |  |  |
| 3 public int sum;                              |         |  |  |
| 4 public int y;                                |         |  |  |
| 5 public Quiz1(){                              |         |  |  |
| 6 y = temp - 1;                                |         |  |  |
| 7 sum = temp + 1;                              |         |  |  |
| 8 temp+=2;                                     |         |  |  |
| 9 }                                            |         |  |  |
| 10 public Quiz1(int p){                        |         |  |  |
| 11 y = temp + p ;                              |         |  |  |
| 12 sum = p + temp + 1;                         |         |  |  |
| 13 temp-=1;                                    |         |  |  |
| 14 }                                           |         |  |  |
| 15 public void methodA(){                      |         |  |  |
| 16 int x=0, y =0;                              |         |  |  |
| 17 y = y + this.y;                             |         |  |  |
| 18 x = this.y + 2 + temp;                      |         |  |  |
| 19 sum = x + y + methodB(x, y);                |         |  |  |
| 20 System.out.println(x + " " + y+ " " + sum); |         |  |  |
| 21 }                                           |         |  |  |
| 22 public int methodB(int m, int n){           |         |  |  |
| 23 int x = 0;                                  |         |  |  |
| 24 y = y + m + (++temp);                       |         |  |  |
| 25 x = x + 2 + n;                              |         |  |  |
| 26 sum = sum + x + y;                          |         |  |  |
| 27 System.out.println(x + " " + y+ " " + sum); |         |  |  |
| 28 return sum;                                 |         |  |  |
| 29 }                                           |         |  |  |
| 30 }                                           |         |  |  |
| 31 public class TestU4{                        |         |  |  |
| 32 public static void main(String [] args){    |         |  |  |
| 33 Quiz1 q1 = new Quiz1();                     |         |  |  |
| 34 q1.methodA();                               |         |  |  |
| 35 q1.methodA();                               |         |  |  |
| 36 Quiz1.temp+= 2;                             |         |  |  |
| 37 Quiz1 q2 = new Quiz1(2);                    |         |  |  |
| 38 q2.methodA();                               |         |  |  |
| 39 q2.methodA();                               |         |  |  |
| 40 }                                           |         |  |  |
| 41 }                                           |         |  |  |